

# Basic Examples and Feature Engineering

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# Goals

- Learn basics of decision tree learning.
- Explore some challenges to using machine learning to mathematical problems.
- Encounter many difficulties, negative results, and arguably trivial results.

# “Good Old Fashioned Machine Learning”

Supervised learning:

- Real world data with inputs (or “features”)  $\mathbf{X}$  and outputs (or “labels”)  $\mathbf{y}$ .
- In practice, split  $\mathbf{X} = \mathbf{X}_{\text{train}} \sqcup \mathbf{X}_{\text{test}}$  and matching  $\mathbf{y} = \mathbf{y}_{\text{train}} \sqcup \mathbf{y}_{\text{test}}$ .
- Learn function  $f$  that “fits”  $f(x) = y$  from the pair  $\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}$ .
- Ideally, for new input  $x$  with unknown output  $y$ ,  $f(x) = y$  (or at least  $|f(x) - y|$  is small).
- Test ideal situation using withheld pair  $\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}}$ .

# Real World Toy Example

Problem: given a Titanic passenger with some information about them (“features”), predict whether or not they survived.

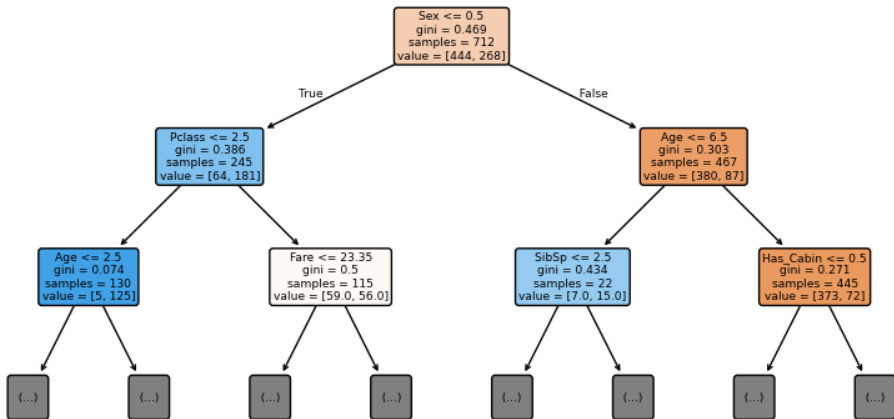
- This data has noise! Impossible to perfectly predict survivability off knowledge of individual passenger available prior to April 15, 1912.
- However, there can still be detectable trends.

We will use data from Kaggle.

| PassengerId | Survived | Pclass | Name | Sex                                               | Age    | SibSp | Parch | Ticket | Fare             | Cabin   | Embarked |   |
|-------------|----------|--------|------|---------------------------------------------------|--------|-------|-------|--------|------------------|---------|----------|---|
| 0           | 1        | 0      | 3    | Braund, Mr. Owen Harris                           | male   | 22.0  | 1     | 0      | A/5 21171        | 7.2500  | NaN      | S |
| 1           | 2        | 1      | 1    | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0  | 1     | 0      | PC 17599         | 71.2833 | C85      | C |
| 2           | 3        | 1      | 3    | Heikkinen, Miss. Laina                            | female | 26.0  | 0     | 0      | STON/O2. 3101282 | 7.9250  | NaN      | S |
| 3           | 4        | 1      | 1    | Futrelle, Mrs. Jacques Heath (Lily May Peel)      | female | 35.0  | 1     | 0      | 113803           | 53.1000 | C123     | S |
| 4           | 5        | 0      | 3    | Allen, Mr. William Henry                          | male   | 35.0  | 0     | 0      | 373450           | 8.0500  | NaN      | S |

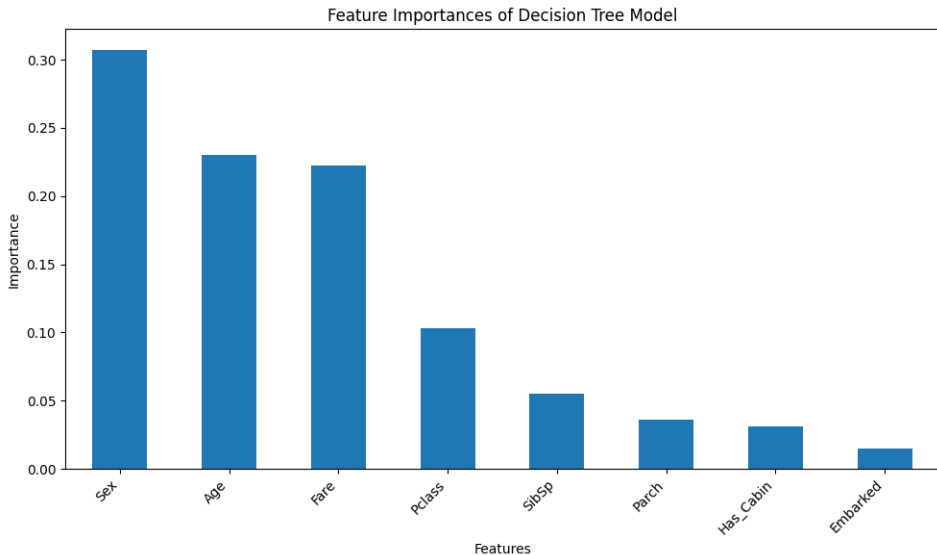
# Decision Trees

Decision Tree Visualization (First 3 Levels)



Accuracy on withheld test data: 79%

# Decision Trees



# Decision Trees

Some pros:

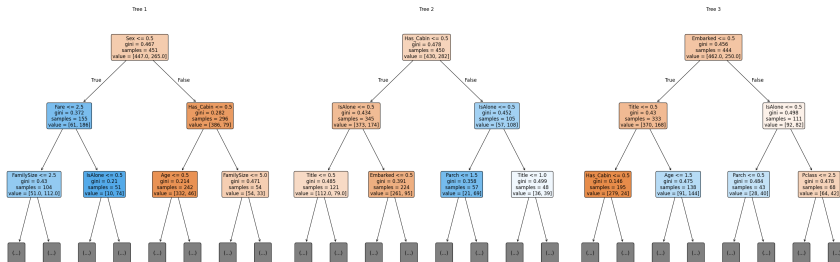
- Relatively easy to understand and interpret.
- Input data requires little preprocessing.

Some cons:

- Highly susceptible to overfitting.
- Cannot detect relationships between features.
- Non-robust: small changes in training data can cause large changes in tree.

# Solution 1: Random Forests

- Instead of training a tree, train a forest!
- For any given classification problem, have every tree vote and take the majority vote.
- Harder to visualize, but can still measure importance of features.



...



## Solution 2: Feature engineering

- If you think there is some relationship between features, you can manually try to add one. (“Derived feature”)
- Titanic data lists “# siblings or spouses” as one feature and “# parents or children” as another. Perhaps total family size is more relevant.
- Titanic data lists every passenger’s name, including their “Title” (e.g., Mr, Master, Miss, Mrs, etc.). This might be useful to extract for the model.

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- Decision tree accuracy with additional features 79%  $\rightarrow$  80%
- Random forest accuracy with additional features 80%  $\rightarrow$  82%

## Solution 3: Gradient boosting

- XGBoost (and other gradient boosted tree libraries) use more advanced techniques to train a decision tree forest in a more sophisticated way to get even better models that are not as likely to overfit.
- Like random forests, some explainability is lost.

# Mathematical Data

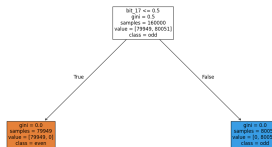
- Unlike real world data, mathematical data (often) has no noise.
- However, decision trees are designed to find signal in noise.
- In general, decision tree learning algorithms are designed for interpolating from data, not extrapolating from data.

## Mathematical Example 1: is this number even or odd?

- E.g., let  $X$  = first  $N$  non-negative integers and  $y_i = 0$  if  $i$  is even, 1 if  $i$  is odd.
- Training using  $X$  as given leads to terrible performance on decision tree.
- Feature engineering: rewrite  $X$  as binary sequences  $\implies$  decision tree model (easily) scores 100%.

# Is this number even or odd? (Binary input)

Decision Tree trained on Binary Parity Data



- Decision tree (binary input)

Weights from all layers of the binary parity model:

```
Layer 0 Weights:
[[-3.3569129e-04  4.2004186e-01]
 [-2.7897890e-04  4.1088238e-01]
 [-4.6315661e-05  -4.2145202e-01]
 [-2.4875102e-04  2.3385565e-01]
 [-2.7676253e-04  -3.3683968e-01]
 [ 4.2838839e-04  -2.5057709e-01]
 [-7.6832675e-04  -2.4937712e-01]
 [ 4.6620599e-04  -5.5159688e-01]
 [ 8.2435130e-05  -2.0094821e-01]
 [-1.4284042e-04  -3.6521530e-01]
 [-3.8786879e-04  -3.8613244e-01]
 [ 7.8174911e-05  6.3660979e-02]
 [ 9.5834243e-05  -2.8157867e-01]
 [-4.0383812e-04  -3.3182439e-01]
 [ 1.2446647e+00  1.6162688e-01]]
```

```
Layer 1 Weights:
[-0.00014593 -0.12783258]
```

```
Layer 2 Weights:
[[2.1112826]
 [0.7268881]]
```

```
Layer 3 Weights:
[-0.72701466]
```

- Neural network (binary input)
- <https://stats.stackexchange.com/questions/161189/train-a-neural-network-to-distinguish-between-even-and-odd-numbers>

## Mathematical Example 2: Horn problem

- Schur polynomials  $s_\lambda(x_1, \dots, x_n)$  form a basis of symmetric polynomials as  $\lambda$  varies over partitions:  
 $\lambda = (\lambda_1 \geq \dots \geq \lambda_n \geq 0) \in \mathbb{Z}_{\geq 0}^n$ .
- Littlewood-Richardson coefficients  $c_{\lambda, \mu}^\nu$ :

$$s_\lambda s_\mu = \sum_{\nu} c_{\lambda, \mu}^\nu s_\nu$$

for  $c_{\lambda, \mu}^\nu \in \mathbb{Z}_{\geq 0}$ .

- Horn problem: determine when  $c_{\lambda, \mu}^\nu \neq 0$  (support).
- Remark: this is a mathematically solved and well understood problem.
- Can we see how ML models could learn the solution?

# Horn problem

Solution (Klyachko, 1998, Knutson-Tao, 1999)

$c_{\lambda\mu}^{\nu} \neq 0 \iff \sum_{i \in I} \lambda_i + \sum_{j \in J} \mu_j \leq \sum_{k \in K} \nu_k$  for  $I, J, K \subseteq \{1, \dots, n\}$   
satisfying  $|I| = |J| = |K|$  and  $|\lambda| + |\mu| = |\nu|$ .



## Additional resource for Algebraic Combinatorics Data

Algebraic Combinatorics Dataset Repository:  
<https://github.com/pnnl/ML4AlgComb>